Reporting-Consistency Audit of 20 JAMA-Family Articles

Synthesis and statistical analysis of retained Major and Minor findings

2026-07-23

Contents

1 Executive summary

This report synthesizes the authoritative final Markdown report in each of 20 article audit packages: 18 `final_report.md` files and two higher-priority `human_adjudication_report.md` files. Only retained items in the final Scientific Findings/Scientific Issues section were counted. Rejected, Uncertain, compliance-screen, and candidate-stage items were excluded.

All 20 articles had at least one retained finding (100%; Wilson 95% CI 83.9%–100.0%). There were 111 findings: 22 Major (19.8%; Wilson 95% CI 13.5%– 28.2%) and 89 Minor (80.2%). 12/20 articles had at least one Major finding (60.0%; Wilson 95% CI 38.7%–78.1%).

Presentation inconsistency was the most frequent formal category (49; 44.1%). Cross-document inconsistency was less common (17) but 8/17 were Major (47.1%), the highest Major density among the main categories. Multi-label artifact coding found supplementary tables in the evidence chain of 69 findings across 18 articles, making them the clearest concentration of risk. Flow diagrams were involved in 15 findings across 10 articles, although only five findings were formally categorized as participant flow inconsistency; the remainder concerned denominators, labels, or cross-document alignment.

> Interpretation: "finding" means retained in the audit report and submitted for human adjudication. It does not necessarily mean author-confirmed error or completed correction.

2 Data and methods

2.1 Inclusion and source selection

- The 20 packages comprise 19 JAMA articles and one JAMA Surgery article; eight package IDs are from 2024 and 12 from 2025.

- A `human_adjudication_report.md` was preferred when present (`jama.2024.19585` and `jama.2024.4183`); otherwise `final_report.md` was used.
- Only final retained scientific findings were parsed. Rejected and Uncertain items were excluded from numerators.
- Each finding has one formal audit category, while artifact locations and text themes are multi-label.
- The complete 111-row ledger is `meta_report/audit_findings.csv`, including the source report path and normalized evidence text.

2.2 Analysis

Counts, proportions, mean, median, quartiles, and range were calculated from the finding-level ledger. Wilson 95% confidence intervals are reported for article-level proportions. Text mining used a rule-based dictionary over each accepted finding's title, locations, compared values, basis, and verification instructions. Thus an artifact/theme count means involvement in the evidence chain, not necessarily the sole root cause, and multi-label totals do not sum to 111.

Each audit could retain no more than 10 final findings; two articles reached that cap. The total of 111 is therefore the count of prioritized final findings, not an uncapped inventory of every detectable defect.

3 Quantitative results

3.1 Burden by article

The mean was 5.55 findings per article (SD 2.87); the median was 5.5, IQR 3–8, range 1–10. The 2024 packages contributed 48 findings across eight articles (mean 6.00; 10 Major), while the 2025 packages contributed 63 across 12 articles (mean 5.25; 12 Major). This is descriptive only: the packages are not a random sample, and year is confounded with package complexity and audit maturity.

Article package	Major	Minor	Total
jama.2024.11057	0	2	2
jama.2024.12829	3	6	9
jama.2024.19585	0	3	3
jama.2024.2302	1	2	3
jama.2024.240147	1	7	8
jama.2024.24764	0	4	4
jama.2024.4183	2	7	9
jama.2024.6063	3	7	10
jama.2025.11178	3	7	10
jama.2025.15440	0	1	1
jama.2025.16450	0	4	4
jama.2025.19563	1	5	6
jama.2025.20765	2	6	8
jama.2025.250116	2	5	7
jama.2025.4390	1	6	7
jama.2025.7583	0	2	2
jama.2025.7710	1	1	2
jama.2025.9110	0	6	6
jama.2025.9663	0	5	5
jamasurg.2025.4929	2	3	5

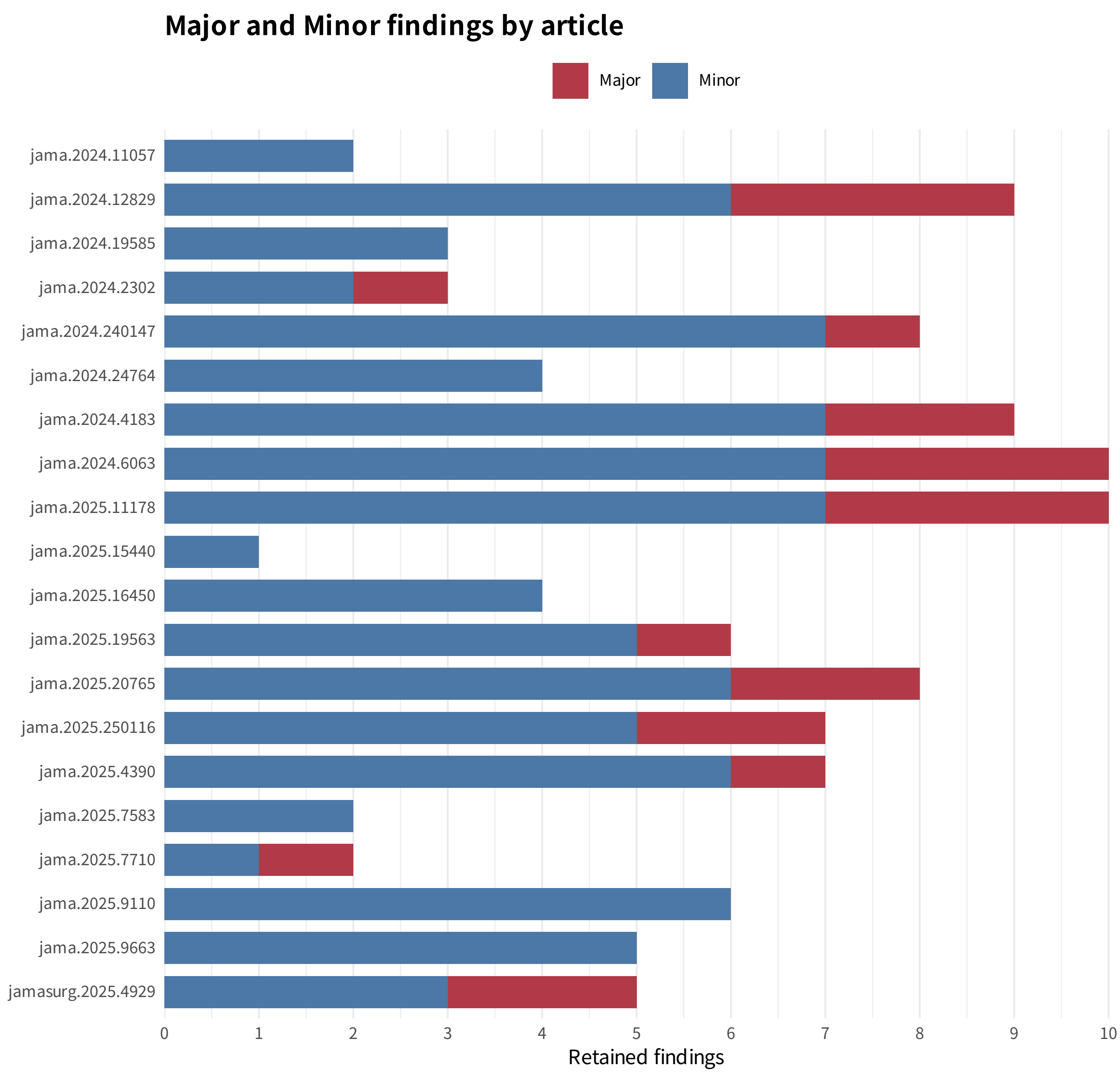

Figure 1: Major and Minor findings by article.

3.2 Formal error categories

Audit category	Findings	Share of 111	Major	Major within category	Articles
Presentation inconsistency	49	44.1%	6	12.2%	19
Statistical reporting inconsistency	24	21.6%	5	20.8%	13
Cross-document inconsistency	17	15.3%	8	47.1%	9
Arithmetic inconsistency	16	14.4%	1	6.2%	9
Participant flow inconsistency	5	4.5%	2	40.0%	4

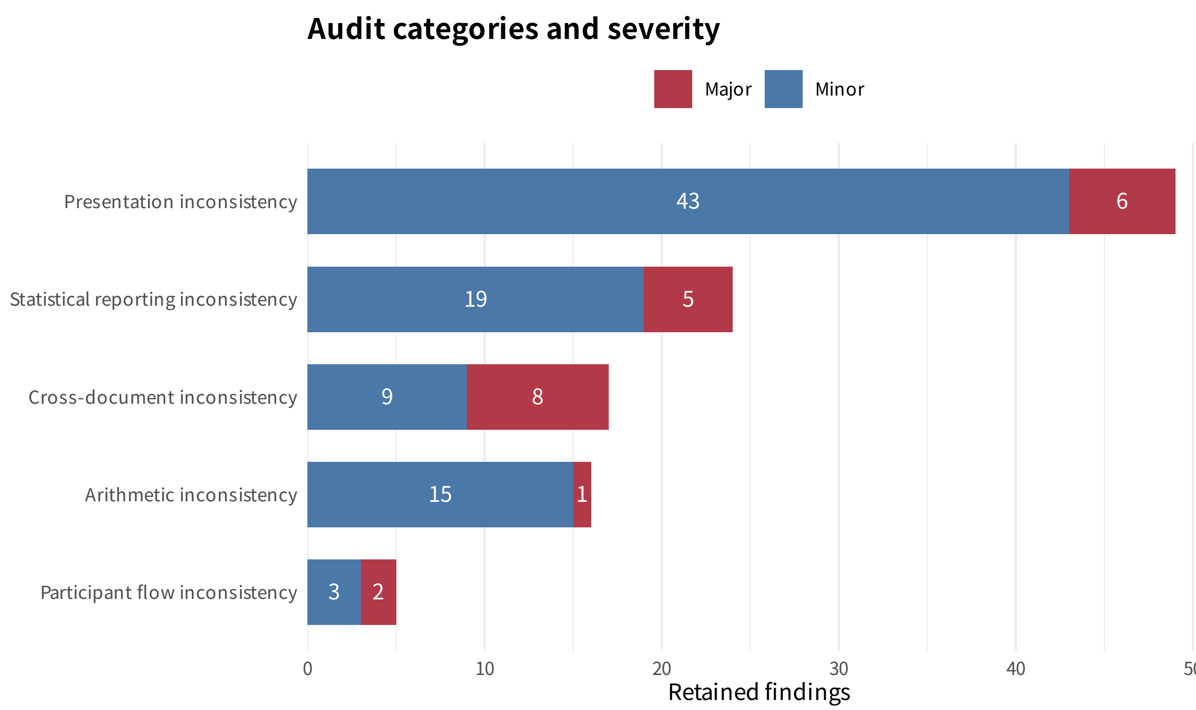

Figure 2: Formal audit category and severity.

Severity structure matters more than frequency alone. Cross-document inconsistency had 8 Major findings among 17, and participant-flow inconsistency had 2 among 5. Presentation findings were far more numerous but only 6/49 were Major; arithmetic inconsistency had only 1 Major among 16. High-frequency low-level defects therefore clustered in presentation, whereas substantive conflict between the main article, supplements, figures, or analysis populations was more likely to become Major.

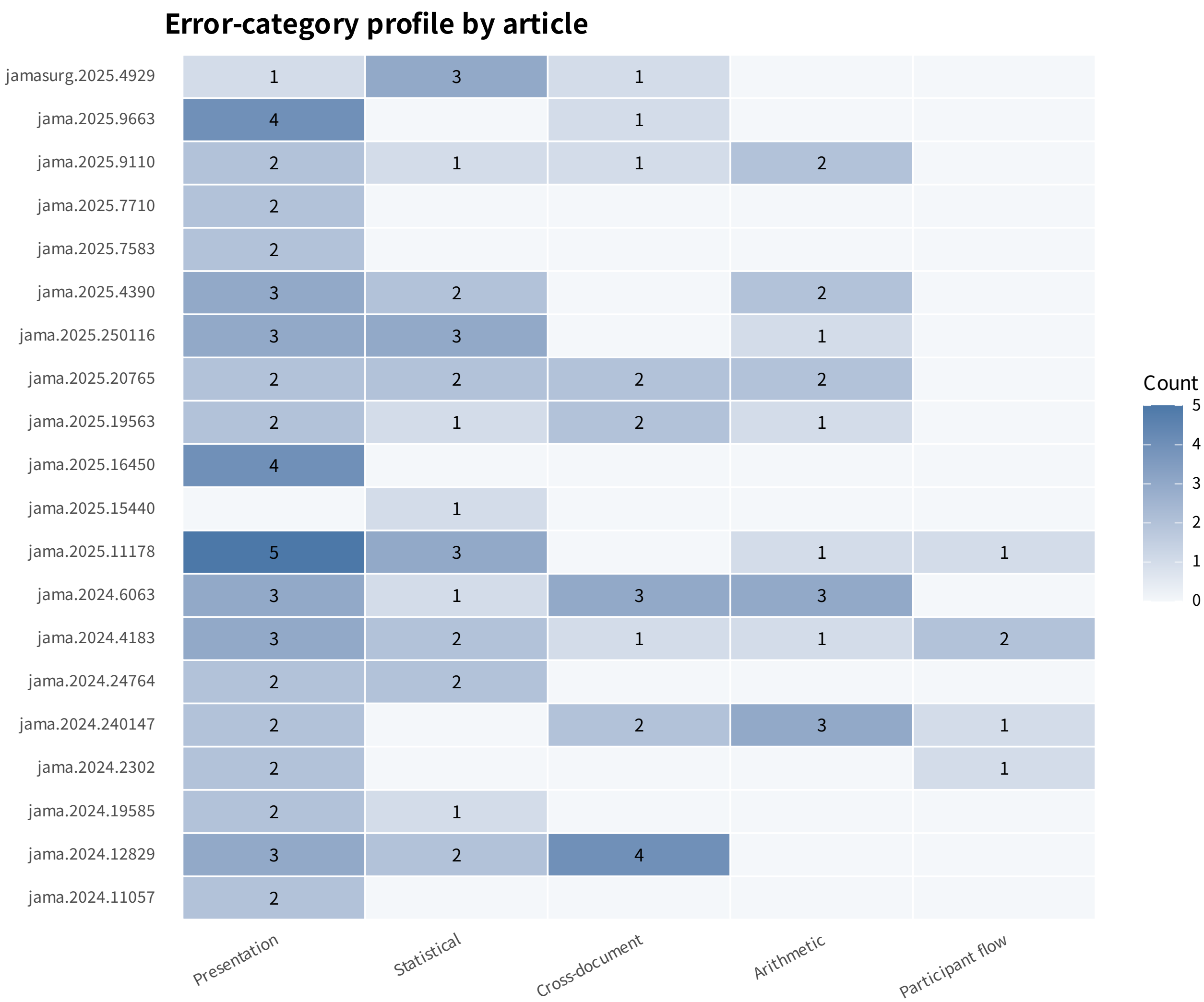

Figure 3: Error-category profile by article.

4 Common failure points

Artifact/location (multi-label)	Finding mentions	Articles
Supplementary table	69	18
Narrative/abstract	28	15
Other figure	22	11
Other main table	21	9
Flow diagram	15	10
Supplementary figure	14	9
Table 1/baseline	9	9

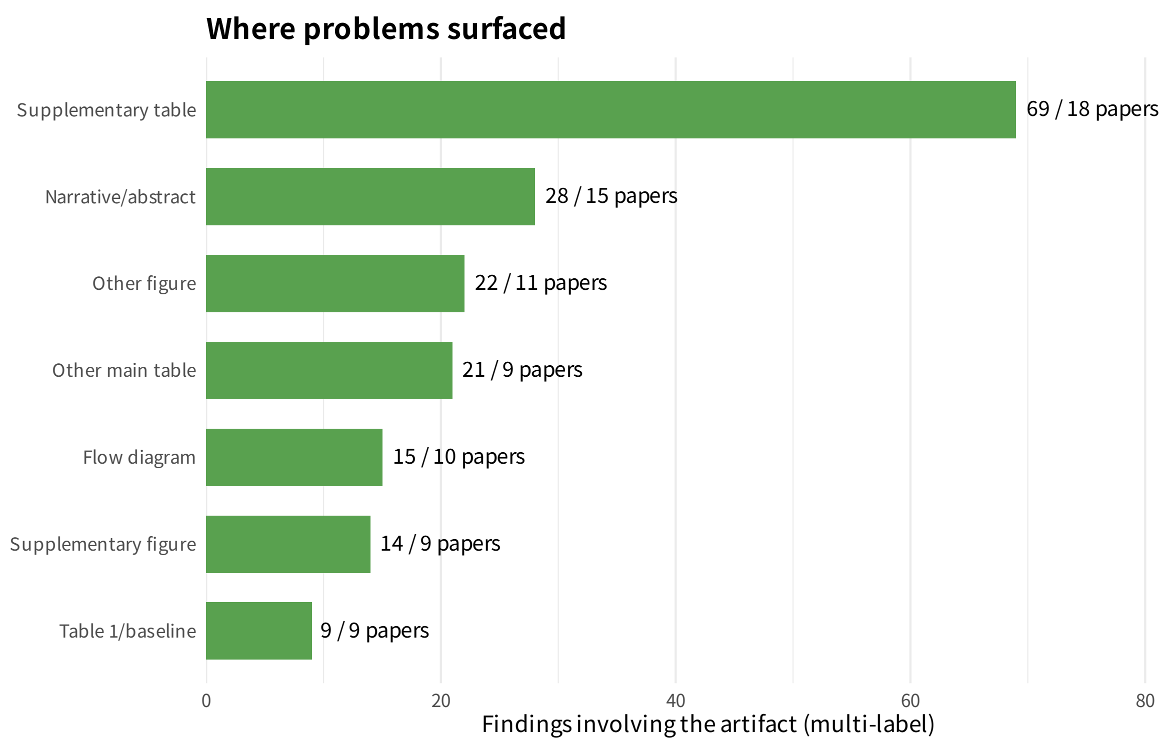

Figure 4: Artifacts involved in each finding's evidence chain; labels are nonexclusive.

4.1 Supplementary tables and Table 1

Supplementary tables were involved in 69/111 findings across 18/20 articles. Recurrent patterns were header denominators that disagreed with cell percentages, mixtures of ITT/PPS/safety populations, duplicated or shifted blocks, mislabeled effect measures or models, and footnotes that contradicted the body. Table 1/baseline material was involved in nine findings across nine articles, including nonconserving sex counts, a one-person FIO2 discrepancy between Table 1 and Figure 2, a repeated calcium-channel blocker percentage error, and different ethnicity-denominator conventions.

4.2 Flow diagrams

Flow/CONSORT diagrams were involved in 15 findings across 10 articles, while only five findings across four articles carried the formal participant-flow category. The difference reflects presentation, population, and cross-document problems that surfaced through a flow figure. High-yield checks are node conservation; explicit separation of randomized, treated, followed, and analyzed populations; postrandomization withdrawal branches; the unit of randomization; and agreement with prose and Table 2 analysis sets.

4.3 Narrative, figures, footnotes, and cross-references

Narrative/abstract text was involved in 28 findings across 15 articles; other main figures in 22 and other main tables in 21. Manual review confirmed at least five direct cross-reference/location errors: `jama.2024.2302`, `jama.2024.6063`, `jama.2025.19563`, `jama.2025.9110`, and `jama.2025.9663`. Dictionary coding also flagged footnotes, titles, captions, and legends repeatedly, showing that semantic labels require checks independent of the numerical cells.

5 Statistical significance, confidence intervals, P values, and definitions

5.1 Confidence intervals

Manual review identified seven findings centered on CI error or incompatibility: `jama.2024.12829`, `jama.2024.4183`, two in `jama.2025.11178`, `jama.2025.15440`, `jama.2025.250116`, and `jamasurg.2025.4929`. Patterns included a point estimate outside its CI, reversed limits, different CIs for the same comparison across abstract and body, and CI/P-value disagreement. The most concentrated example was `jama.2025.11178`: seven rows had 95% CIs excluding zero while their P values exceeded .05, plus a separate group of SMDs outside their CIs.

5.2 P values and test methods

At least five findings made the P value or test method a central contradiction: "Chi-square" rows that reproduced Fisher exact results (`jama.2024.12829`); an unqualified no-difference narrative despite day-7 P=.02 and a CI excluding zero (`jama.2024.24764`); grouped CI–P disagreement (`jama.2025.11178`); two age P-value footnotes (`jama.2025.19563`); and identical binary counts with different P values (`jama.2025.4390`). These should be separated into calculation errors, method-label errors, and narrative failure to distinguish global interaction from time-specific contrasts.

5.3 Text and conceptual definitions

Recurring conceptual defects included an OR labeled "difference," RR expanded as risk difference, a univariate estimate described as multivariable, within-stratum treatment ORs labeled interactions, median(IQR) labeled mean(SD), FiO2 fractions labeled percentages, and one analysis unit called women, infants, and patients. These defects may leave cell values unchanged while materially changing interpretation of the estimand, population, or model.

Dictionary theme (multi-label)	Finding hits	Articles
Count/percentage arithmetic	72	20
Denominator/population	54	19
Footnote/title/caption/legend	45	17
P value/test labeling	23	11
Participant-flow wording	21	13
Summary-statistic/unit label	20	13
Duplicate/transposed content	20	13

Dictionary theme (multi-label)	Finding hits	Articles
Effect measure/model label	20	12
CI/point-estimate compatibility	18	13
Cross-reference/citation	5	5
Other wording/definition	3	2

6 The 22 Major findings

Article	Finding	Category	Major finding synopsis
jama.2024.12829	1	Statistical reporting inconsistency	Stroke incidence-difference CI excludes its point estimate
jama.2024.12829	2	Cross-document inconsistency	Figure S5 conflicts with its denominator and Table S11 disabling-stroke counts
jama.2024.12829	3	Cross-document inconsistency	Table S6 mixes ITT and PPS data
jama.2024.2302	C1	Presentation inconsistency	Abstract mislabels the 320 postwithdrawal cohort as repaired; actual repair total is 281
jama.2024.240147	F1	Participant flow inconsistency	Randomized, intervention-stage, and analyzed populations are conflated
jama.2024.4183	C04	Cross-document inconsistency	CNRT+ lozenge dose conflicts as 2 mg versus 4 mg
jama.2024.4183	C08	Presentation inconsistency	Two supplemental estimates/intervals appear decimal-shifted
jama.2024.6063	C04	Arithmetic inconsistency	eTable 4 pain-change block duplicates an unrelated function row and conflicts with endpoints
jama.2024.6063	C05	Presentation inconsistency	Lower-leg-strength inference block exactly duplicates a back-pain row
jama.2024.6063	C07	Cross-document inconsistency	Main Table and eTable 7 adverse-event totals differ by five
jama.2025.11178	1	Participant flow inconsistency	Follow-up-pattern cells exceed randomized and followed totals
jama.2025.11178	2	Statistical reporting inconsistency	Seven 95% CIs exclude zero while paired P values exceed .05

Article	Finding	Category	Major finding synopsis
jama.2025.11178	4	Statistical reporting inconsistency	Multiple SMDs fall outside CIs, have reversed endpoints, or opposite signs
jama.2025.19563	C-06	Cross-document inconsistency	Figure 3B uses a larger HbA1c analysis population than eTable 14 without explanation
jama.2025.20765	F01	Cross-document inconsistency	eTable 2 omits a 40-participant mHealth cluster
jama.2025.20765	F02	Cross-document inconsistency	Prose reverses the direction of irritability and anxiety events
jama.2025.250116	C04	Presentation inconsistency	eFigure 8 repeats eFigure 7 inference values despite different event cells
jama.2025.250116	C05	Statistical reporting inconsistency	High-stratum treatment ORs are labeled and interpreted as interactions
jama.2025.4390	SCI-01	Presentation inconsistency	Figure 3 rate columns contain person-time
jama.2025.7710	1	Presentation inconsistency	Primary-outcome analysis unit conflicts among women, infants, and patients
jamasurg.2025.4929	F-01	Statistical reporting inconsistency	pN N3 row is internally incompatible across counts, percentage, OR, CI, and P value
jamasurg.2025.4929	F-04	Cross-document inconsistency	Main text presents a univariate approach estimate as multivariable

7 Article-by-article synthesis (Major and Minor)

7.1 jama.2024.11057

Major 0; Minor 2; total 2.

Two Minor findings: eTable 4 labels mean(SD)-like values as median(IQR), and the eTable 5 responder-only title also covers randomized-denominator rows.

7.2 jama.2024.12829

Major 3; Minor 6; total 9.

Three Major findings: a stroke-difference CI excludes its point estimate; Figure S5 conflicts on denominator/disabling stroke counts; and Table S6 mixes ITT and PPS data. Six Minor findings concern S7–S9 headers/denominators, center counts, a complication count, and test-method labeling.

7.3 jama.2024.19585

Major 0; Minor 3; total 3.

Three Minor findings: eTable 10 labels an OR as a difference, Figure 2 does not state time-specific denominators, and the eFigure 3 legend fails to define displayed P values.

7.4 jama.2024.2302

Major 1; Minor 2; total 3.

One Major finding: the abstract says 320 infants underwent repair, whereas the article and flow diagram total 281. Two Minor findings concern an omitted withdrawal branch and an enrollment cross-reference that should point only to eTable 1.

7.5 jama.2024.240147

Major 1; Minor 7; total 8.

One Major finding: the abstract/Key Points call analyzed or intervention-stage populations randomized. Seven Minor findings include center percentages, 4/743 and 0/747 errors, an orphan footnote, cross-display percentages, and an undisclosed denominator.

7.6 jama.2024.24764

Major 0; Minor 4; total 4.

Four Minor findings: undisclosed multiple responses, an omitted UK-only subgroup restriction, missing subgroup denominators, and a quality-of-life narrative inconsistent with the day-7 CI/P value.

7.7 jama.2024.4183

Major 2; Minor 7; total 9.

Two Major findings: the CNRT+ lozenge dose conflicts across documents, and two supplemental values appear decimal-shifted. Seven Minor findings involve flow arithmetic/wording, sex counts, direction, power monotonicity, denominators, and adverse-event qualification.

7.8 jama.2024.6063

Major 3; Minor 7; total 10.

Three Major findings: a copied/incorrect change block, an unrelated row duplicated wholesale, and conflicting adverse-event totals. Seven Minor findings concern estimate sign, percentages, arm reversals, adherence, wrong eTable citations, and layout.

7.9 jama.2025.11178

Major 3; Minor 7; total 10.

Three Major findings: follow-up-pattern counts do not conserve N; seven CIs exclude zero while P>.05; and SMD estimates/CIs/signs are impossible. Seven Minor findings concern percentages, duplicate or missing labels, subgroup headers, adjustment status, and mean/median wording.

7.10 jama.2025.15440

Major 0; Minor 1; total 1.

One Minor finding: the same stroke comparison has different 95% CIs in the abstract and the Results/Figure 4B.

7.11 jama.2025.16450

Major 0; Minor 4; total 4.

Four Minor findings: undisclosed GDB denominators, a B+S header mismatch, a one-person FIO2 discrepancy between Table 1 and Figure 2, and RR expanded as risk difference.

7.12 jama.2025.19563

Major 1; Minor 5; total 6.

One Major finding: Figure 3B visibly uses a larger HbA1c population than eTable 14 without explanation. Five Minor findings concern 10/59, three wrong eTable numbers, age-P-value footnotes, BMI called weight, and percent versus percentage points.

7.13 jama.2025.20765

Major 2; Minor 6; total 8.

Two Major findings: a 40-person mHealth cluster is omitted and adverse-event direction is reversed in prose. Six Minor findings cover death/adverse-event percentages, an unidentified population, death-excluded ITT wording, and a title/body mismatch.

7.14 jama.2025.250116

Major 2; Minor 5; total 7.

Two Major findings: eFigure 8 repeats eFigure 7 inference values, and high-stratum treatment ORs are labeled interactions. Five Minor findings concern a narrative count, OR decimal error, a point estimate outside its CI, and duplicated/missing table characters.

7.15 jama.2025.4390

Major 1; Minor 6; total 7.

One Major finding: Figure 3 columns labeled rates per 100 patient-years contain about 71 hundreds of patient-years. Six Minor findings include a duplicated ethnicity row, adjusted/unadjusted CI labeling, identical counts with different P values, geography totals, adherence, and a repeated percentage error.

7.16 jama.2025.7583

Major 0; Minor 2; total 2.

Two Minor findings: inclusion wording is applied to excluded patients, and the MAGIC-MT control event count is missing.

7.17 jama.2025.7710

Major 1; Minor 1; total 2.

One Major finding: the same primary-outcome analysis unit is called women, infants, and patients. One Minor finding: prose and Table 1 use different placebo-ethnicity denominator conventions.

7.18 jama.2025.9110

Major 0; Minor 6; total 6.

Six Minor findings: sex counts, a protocol-deviation percentage, patient-versus-ICU randomization wording, two mean(SD)/median(IQR) labels, and a Supplement 1/3 locator.

7.19 jama.2025.9663

Major 0; Minor 5; total 5.

Five Minor findings: a wrong eFigure citation, mixed analytic subsets in a summary, a truncated year, FiO2 fractions labeled percentages, and an undefined asterisk.

7.20 jamasurg.2025.4929

Major 2; Minor 3; total 5.

Two Major findings: the pN N3 regression row is internally impossible, and a univariate estimate is presented as multivariable. Three Minor findings concern the age row, five unreproducible ORs, and a CONSORT refusal label.

8 Interpretation, limitations, and recommendations

1. **Audit supplementary tables first.** Apply column-level denominator inference, percentage recomputation, estimate/CI/P-value checks, and adjacent-row duplicate block detection.
2. **Use an explicit population-stage dictionary.** Randomized, treated, safety, ITT, complete-case, PPS, and death-excluded populations should never rely on context alone.
3. **Validate figures back against generation data.** Priorities are flow conservation, figure units and headers, eFigure copy/paste, and caption definitions.
4. **Automate cross-reference resolution.** Every eTable/eFigure/Supplement locator in the main article should map one-to-one to the actual title and file location.
5. **Make statistical consistency a release gate.** A point estimate must lie within its CI; two-sided P values, CIs, and null conclusions should align; identical 2×2 cells under the same method cannot yield different P values; adjustment and model labels must agree with source output.

Limitations include the 10-finding cap, nonrandom package selection, heterogeneous document volume/OCR/audit scope, use of the source audit taxonomy, and rule-based multi-label text coding. Retained/accepted findings were submitted for or framed as human adjudication and should not be read as completed author-confirmed errata.

9 Reproducibility and provenance

- `meta_report/audit_findings.csv`: 111-row finding ledger.
- `meta_report/article_counts.csv`: Major/Minor counts by article.
- `meta_report/category_counts.csv`: category-by-severity counts.
- `meta_report/artifact_counts.csv`: multi-label artifact counts.
- `meta_report/mechanism_counts.csv`: multi-label text-theme counts.
- `meta_report/generate_report.py` and `meta_report/analysis.R`: reproducible scripts.

Authoritative source reports:

- `jama.2024.11057`: `jama.2024.11057/.ai_paper_validation/final_report.md`
- `jama.2024.12829`: `jama.2024.12829/.ai_paper_validation/final_report.md`
- `jama.2024.19585`: `jama.2024.19585/.ai_paper_validation/human_adjudication_report.md`
- `jama.2024.2302`: `jama.2024.2302/.ai_paper_validation/final_report.md`
- `jama.2024.240147`: `jama.2024.240147/.ai_paper_validation/final_report.md`
- `jama.2024.24764`: `jama.2024.24764/.ai_paper_validation/final_report.md`
- `jama.2024.4183`: `jama.2024.4183/.ai_paper_validation/human_adjudication_report.md`
- `jama.2024.6063`: `jama.2024.6063/.ai_paper_validation/final_report.md`
- `jama.2025.11178`: `jama.2025.11178/.ai_paper_validation/final_report.md`
- `jama.2025.15440`: `jama.2025.15440/.ai_paper_validation/final_report.md`
- `jama.2025.16450`: `jama.2025.16450/.ai_paper_validation/final_report.md`
- `jama.2025.19563`: `jama.2025.19563/.ai_paper_validation/final_report.md`
- `jama.2025.20765`: `jama.2025.20765/.ai_paper_validation/final_report.md`
- `jama.2025.250116`: `jama.2025.250116/.ai_paper_validation/final_report.md`
- `jama.2025.4390`: `jama.2025.4390/.ai_paper_validation/final_report.md`
- `jama.2025.7583`: `jama.2025.7583/.ai_paper_validation/final_report.md`
- `jama.2025.7710`: `jama.2025.7710/.ai_paper_validation/final_report.md`
- `jama.2025.9110`: `jama.2025.9110/.ai_paper_validation/final_report.md`
- `jama.2025.9663`: `jama.2025.9663/.ai_paper_validation/final_report.md`
- `jamasurg.2025.4929`: `jamasurg.2025.4929/.ai_paper_validation/final_report.md`